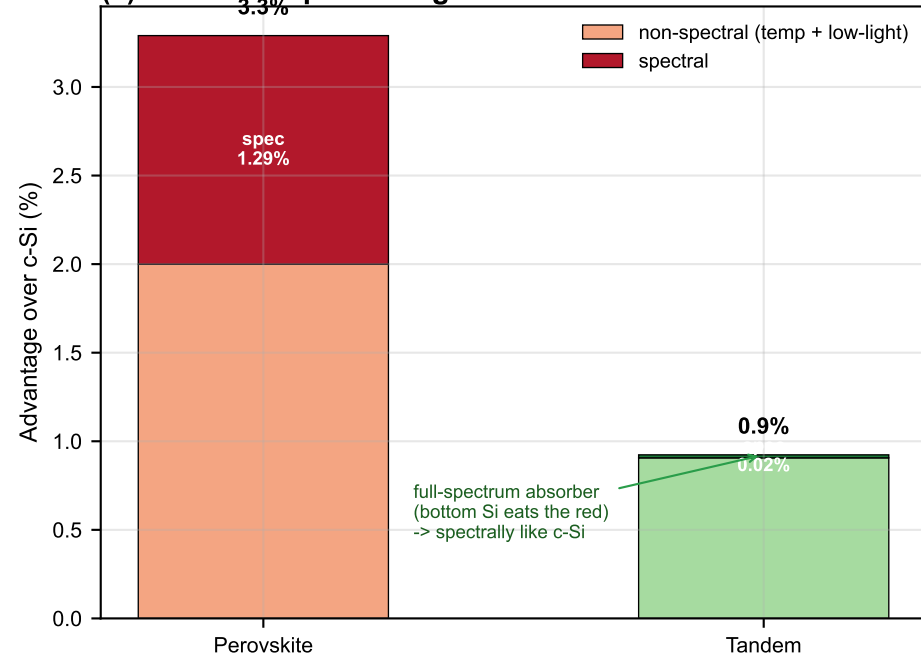
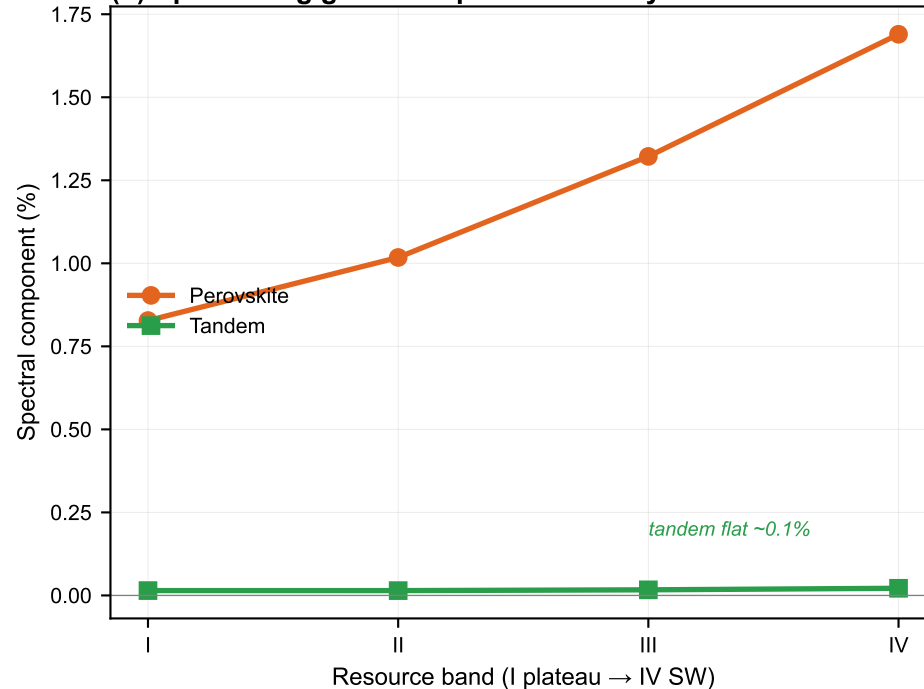


Against modern silicon the tandem has almost no per-kWp edge (0.9%) and no spectral component — a full-spectrum absorber; its value is area, not utility yield

(a) Tandem's spectral leg ~ 0



(b) Spectral leg grows for perovskite only



(c) Only perovskite tracks the spectrum

